

智能科学与技术专业培养方案

Curriculum for Undergraduate of Intelligent Science and Technology Major

一、专业介绍

本专业从 2024 年开始招收本科生，属于工学一级学科。本专业为学校重点建设专业，是人工智能、机器人、智能系统等多学科交叉融合的基础性本科专业。定位为培育“科班出身的人工智能专家”，旨在培养既具备计算机科学与技术知识，同时也具备人工智能、机器学习、深度学习等前沿技术应用能力的高级人才。

I. Professional Introduction

Starting from 2024, this major will admit undergraduate students and is classified as a first-level discipline in engineering. As a key construction major of the university, it is a fundamental undergraduate program that integrates multiple disciplines such as artificial intelligence, robotics, and intelligent systems. The program aims to cultivate “AI experts with a scientific background” by providing students with a solid foundation in computer science and technology, as well as advanced capabilities in the application of cutting-edge technologies such as artificial intelligence, machine learning, and deep learning.

二、培养目标

本专业围绕学校“钢铁品质、社会英才”的人才培养总目标，培养适应国民经济发展需求，面向工业与信息化建设，德智体美劳全面发展的社会主义事业合格建设者和可靠接班人，具有良好的人文社会科学素养、社会责任感、职业道德、国际视野、创新精神，能够将智能科学与技术领域的基础理论和专门知识用于解决复杂工程问题，能够在人工智能、机器人、互联网、信息产业、钢铁冶金等行业的智能科学与技术相关领域部门中，从事智能系统的研究、设计、开发、工程管理等工作的拔尖创新型人才。

本专业毕业生工作 5 年左右预期达到的职业能力如下：

1. 积极投身社会主义建设事业，自觉践行社会主义核心价值观，具有良好的人文素养、社会责任感，遵守职业道德和规范；
2. 能够将数学、自然科学和专业知知识用于解决在智能科学与技术及相关领域遇到的复杂工程问题；
3. 能够设计针对智能科学与技术领域复杂工程问题的解决方案，能够在智能科学与技术相关领域的复杂工程项目中胜任研究、设计、开发和项目管理工作；
4. 具有一定的国际视野，能在多学科、多文化背景下与业界同行及社会公众进行有效沟通和交流，能够胜任团队的核心或领导工作；
5. 能够适应国民经济发展需求、不断地自主学习，在工作中不断创新。

II. Training objectives

This major focuses on the general goal of cultivating talents of "steel quality, social talents" of the university, and aims to cultivate students with good humanistic quality, international vision, innovative spirit, solid theoretical knowledge and professional skills of intelligent science and technology, strong team cooperation ability and lifelong learning ability, to solve complex engineering problems. Make it become a qualified builder and reliable successor of the socialist cause of all-round development of morality, intelligence, physique, beauty and labor. Top innovative talents who are capable of working in related departments of artificial intelligence, robotics, internet, information industry, iron and steel metallurgy, engaged in the design and development of intelligent hardware and software systems, system testing, operation and maintenance, project management and other works.

Graduates of this major are expected to achieve the following professional abilities in about 5 years of work:

1. Have a good human arts literacy, social responsibility, adhere to professional ethics and norms, and be able to consider the impact of social, health, safety, legal, cultural and environmental factors in the practice of engineering projects in computer and related fields.
2. Be able to use mathematics, natural science and professional knowledge to solve complex engineering problems in intelligent science and technology and its

related fields in software and hardware system research and development, analysis and design, development and testing, project management and other engineering practices.

3. Capable of designing solutions for complex engineering problems in the field of intelligent science and technology, also capable of conducting research, design, development, and project management in complex engineering projects related to artificial intelligence.

4. Good expression, communication, coordination and leadership skills as well as foreign language application skills; able to organize and manage project teams in this field in a multi-disciplinary and multi-cultural context.

5. With a good international vision and lifelong learning ability, they can actively adapt to the changes of social development and career development demands, constantly update my professional knowledge and improve my personal ability and technical level through independent learning.

三、专业特色

依托“智能信息处理与实时工业系统”湖北省重点实验室。拥有一支具有国际视野的高学历、高水平教师队伍为主，教师 100%具有博士学位，知识水平和层次较高。本专业以计算机知识为基础、以智能机器人技术为特色、以协作机器人和复合机器人为平台进行学生培养。结合教师的科研

成果和师资基础，开设自然语言处理、大数据技术原理与应用、计算机视觉、机器人理论与技术等多门具有专业特色的课程及实践。

III. Professional characteristics

Relying on the Hubei Province Key Laboratory of Intelligent Information Processing and Real-time Industrial System, this major boasts a highly educated and high-level faculty team with an international perspective, where all teachers hold a doctoral degree, demonstrating a high level of knowledge and expertise. With computer knowledge as its foundation, this major features intelligent robot technology and uses collaborative robots and composite robots as platforms for student training. Leveraging the research achievements of the faculty and their expertise, the program offers a range of courses and practical training in specialized areas such as natural language processing, big data technology and applications, computer vision, and robot theory and technology.

四、毕业要求

- 1.工程知识：具备数学、自然科学知识，系统掌握计算机领域的工程基础和专业基础知识，能够将相关知识用于解决计算机领域的复杂工程问题。
- 2.问题分析：能够综合运用数学、自然科学和工程科学的基本原理和方法，通过文献研究，对计算机领域的复杂工程问题进行识别、表达和分析，以获得有效结论。
- 3.设计/开发解决方案：能够综合运用理论和技术手段，针对计算机领域的复杂工程问题提出解决方案，设计满足特定需求的计算机软硬件系统、模

块或开发流程，够在设计中体现创新意识，并能够综合考虑社会、健康、安全、法律、文化以及环境等制约因素。

4.研究：能够基于科学原理并采用科学方法对计算机领域的复杂工程问题进行研究，制定技术路线、设计实施方案、分析解释实验数据，通过信息综合得到合理有效的结论。

5.使用现代工具：能够针对计算机领域的复杂工程问题，开发、选择与使用恰当的软硬件开发环境与工具、信息检索与分析工具，对复杂计算机工程问题进行仿真与模拟，并能够理解其局限性。

6.工程与可持续发展：能够基于计算机工程相关背景知识进行合理分析，评价计算机领域软硬件开发、系统设计等工程实践和复杂工程问题解决方案对环境、社会可持续发展的影响。并理解应承担的社会责任。

7. 工程伦理和职业规范。有工程报国、为民造福的意识，具有良好的人文社会科学素养和较强的社会责任感，树立和践行社会主义核心价值观，能够在计算机软硬件系统设计与开发等工程实践中理解并遵守工程职业道德和规范，履行计算机工程师的社会责任。

8.个人与团队：具有较强的团队合作意识与能力，能够正确理在多学科背景下的团队中承担个体、团队成员以及负责人的角色，并承担相应责任和义务。

9. 沟通：能够就计算机领域的复杂工程问题与业界同行及社会公众进行有效沟通和交流，能够撰写报告和设计文稿、陈述发言、清晰表达或回应指令，并具备一定的国际视野，能够在跨文化背景下进行沟通和交流。

10. 项目管理：理解并掌握计算机工程项目的成本、进度、范围、质量、风险等管理原理与经济决策方法，并能在多学科环境中应用。

11. 终身学习：具有自主学习和终身学习的意识，能够追踪计算机专业领域的发展动态，有不断学习和适应发展的能力。

IV. Requirements

1. Engineering knowledge: I have the knowledge of mathematics and natural science, have a systematic grasp of the engineering foundation and professional knowledge in the field of computer, and can apply the relevant knowledge to solve complex engineering problems in the field of computer.

2. Problem analysis: I can use the basic principles and methods of mathematics, natural science and engineering science comprehensively to identify, express and analyze complex engineering problems in the field of computer through literature research, so as to obtain effective conclusions.

3. Design/Development solutions: can the integrated use of theory and technology, in view of the complex engineering problems put forward solutions in the field of computer, designed to meet the specific needs of the computer hardware and software system, module, or the development process, is reflected in the design innovation, and to consider the social, health, safety, legal, cultural, and environmental factors, etc.

4. Research: I can study complex engineering problems in computer field based on scientific principles and using scientific methods, formulate technical routes,

design implementation plans, analyze and interpret experimental data, and draw reasonable and effective conclusions through information synthesis.

5. Use modern tools: Able to develop, select and use appropriate software and hardware development environments and tools, information retrieval and analysis tools, simulate complex computer engineering problems, and understand their limitations for complex engineering problems in the computer field.

6. Engineering and sustainable development: Be able to conduct reasonable analysis based on relevant background knowledge of computer engineering, evaluate the impact of engineering practices such as software and hardware development and system design in computer field and solutions to complex engineering problems on the sustainable development of the environment and society, and understand the social responsibilities that should be undertaken.

7. Ethics and Professional norms: have good humanistic and social science literacy and strong social responsibility, establish and practice the socialist core values, be able to understand and abide by the engineering professional ethics and norms in the design and development of computer software and hardware systems and other engineering practices, and fulfill the social responsibility of computer engineers.

8. Individual and team: Strong sense and ability of teamwork, able to correctly assume the roles of individual, team member and leader in a multi-disciplinary team, and assume corresponding responsibilities and obligations.

9. Communication: able to effectively communicate and communicate with industry peers and the public on complex engineering issues in the field of computer, able to write reports and design documents, make presentations, clearly express or respond to instructions, and have a certain international perspective and be able to communicate and communicate in a cross-cultural context.

10. Project Management: Understand and master the management principles and economic decision-making methods of computer engineering projects, such as cost, schedule, scope, quality and risk, and be able to apply them in a multidisciplinary environment.

11. Lifelong learning: have the awareness of self-learning and lifelong learning, can track the development of the computer field, have the ability to constantly learn and adapt to the development.

附：培养目标实现矩阵

	培养目标 1	培养目标 2	培养目标 3	培养目标 4	培养目标 5
毕业要求 1		√			
毕业要求 2		√	√		
毕业要求 3		√	√		√
毕业要求 4		√	√		
毕业要求 5		√	√		
毕业要求 6	√				
毕业要求 7	√			√	
毕业要求 8				√	√
毕业要求 9				√	√
毕业要求 10			√	√	
毕业要求 11	√				√

五、专业主干课程

面向对象程序设计，离散数学，数据库系统原理，数字逻辑与数字系统，计算机组成原理，数据结构，操作系统，软件工程，算法设计与分析，计算机网络，机器学习，计算机视觉，人工智能，自然语言处理，机器人理论与技术。

V. Core courses

Object-Oriented Programming, Discrete Mathematics, Principles of Database System, Digital Logic and Digital System, Principles of Computer Organization, Data Structure, Operating System, Software Engineering, Algorithm Design and Analysis, Computer Network, Machine Learning, Computer Vision, Artificial Intelligence, Natural Language Processing, Robotic Theory and Technology.

六、基本学制：四年

VI. Recommended length of the program: 4 years

七、授予学位：工学学士

VII. Degree: Bachelor of Engineering

学生修满所规定的最低毕业学分，符合武汉科技大学授予学士学位规定，授予工学学士学位。

八、毕业学分要求：160 学分

课程类型		学分要求	课程类型	学分要求	
1、公共课程平台		42	3、专业课程模块	45.5	
公共基础课程		28	专业必修课程	41.5	
通识教育课程	必修	8	专业选修课程	4	
	选修	6	4、实践教学模块	34	
2、学科基础平台		32.5	大类实践课程	3	
大类学科基础课程	必修	9	专业实践课程	必修	31
	选修	0		选修	0
专业学科基础课程		23.5	5、素质拓展模块	6	

VI、毕业学分要求：160 学分

Type of courses		Academic credits	Type of courses		Academic credits
1.Common Courses		42	3. Specialized Courses		45.5
Common Basic Courses		28	Required Courses		41.5
General Education Courses	Required Courses	8	Elective Courses		4
	Elective Courses	6	4.Practicum and Internship Courses		34
2.General Disciplinary Courses		32.5	Large Category Practical Courses		3
Large Category Basic Courses	Required Courses	9	Disciplinary Practical Courses	Required Courses	31
	Elective Courses	0		Elective Courses	0
Disciplinary Basic Courses		23.5	5. Quality Development Courses		6

九、学分比例

VII、Ratio of Credits

1. 必修选修学分比例

The proportion of compulsory elective credits

类别	学分	占总学分比例
必修	150	93.75%
选修	10	6.25%

2. 实践教学环节学分比例

The proportion of credits in practice teaching

类别	学分	占总学分比例
实践教学环节	实验教学学时	13.9
	实践教学模块	34
	素质拓展模块	6
		32.438%

八、毕业要求实现矩阵

VIII. Graduation Realization Matrix

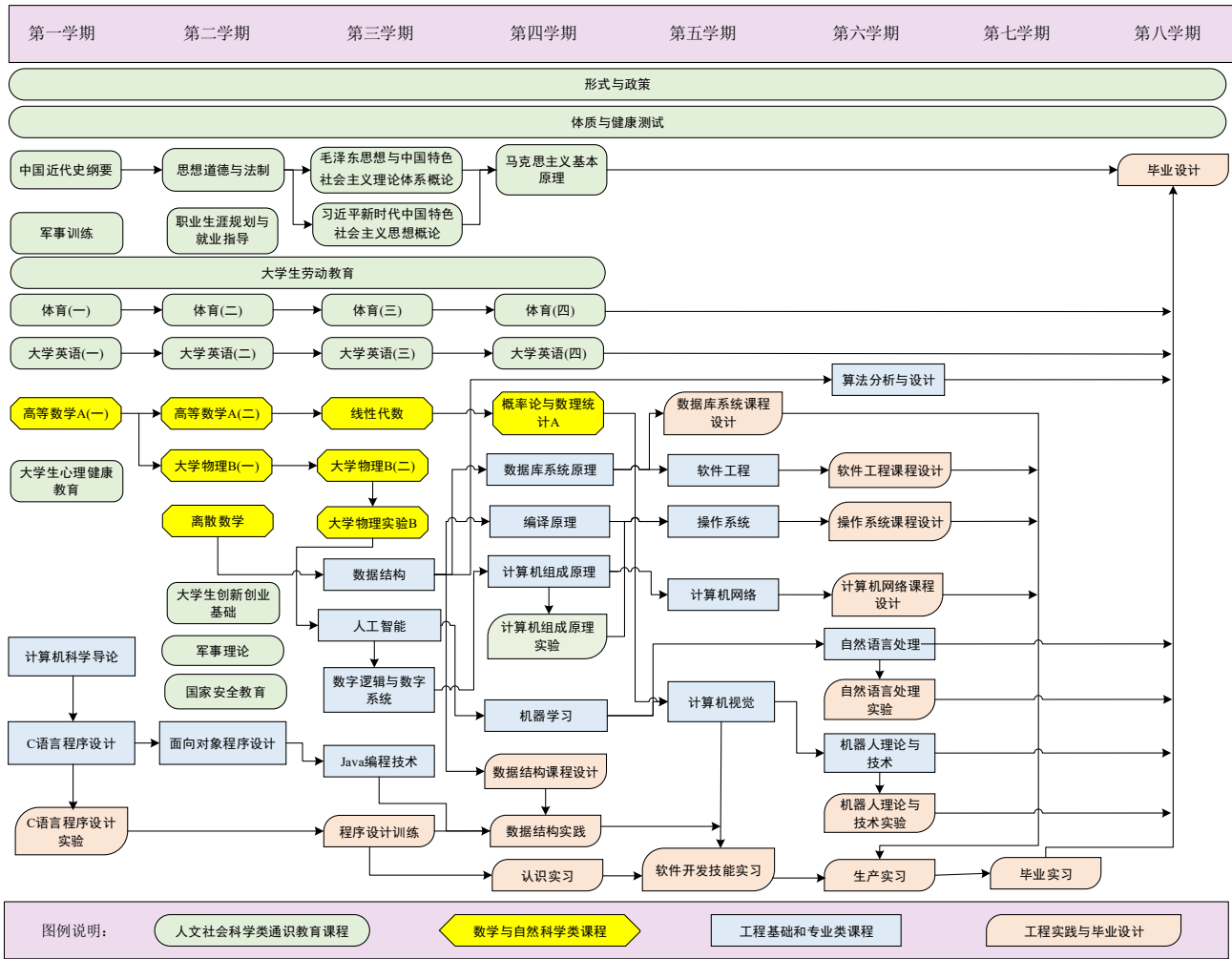
课程名称	智能科学与技术专业毕业要求										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
思想道德与法治			√			√	√				
中国近现代史纲要							√				√
马克思主义基本原理							√				√
毛泽东思想和中国特色社会主义理论体系概论							√				
习近平新时代中国特色社会主义思想概论							√				
形势与政策						√					√
大学英语									√		√

课程名称	智能科学与技术专业毕业要求										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
体育								√			
军事理论							√	√			
军事训练								√			
大学生心理健康教育							√				√
职业生涯规划与就业指导							√	√			√
大学生创新创业基础						√	√				
高等数学 A	√	√									
大学物理 B	√	√									
大学物理实验 B		√		√							
线性代数	√	√									
概率论与数理统计 A	√			√							
机器学习	√	√		√							
离散数学	√	√									
C 语言程序设计	√		√		√						
面向对象程序设计		√	√		√						
Java 编程技术			√	√							
数据结构	√	√	√								
计算机组成原理		√	√								
编译原理	√	√		√							
操作系统		√	√	√							
数据库系统原理	√	√	√								
计算机网络	√	√		√							
软件工程			√	√						√	
计算机科学导论	√					√					√
数字逻辑与数字系统	√	√									
自然语言处理				√			√				√

课程名称	智能科学与技术专业毕业要求										
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
机器人理论与技术				√		√					√
人工智能		√	√		√						
算法设计与分析	√		√	√							
计算机视觉				√	√				√		√
C 语言程序设计实验				√	√		√				
计算机组成原理实验				√	√				√		
数据结构课程设计		√		√					√		
数据库系统课程设计		√	√						√		
操作系统课程设计			√						√		
计算机网络课程设计			√	√		√					
自然语言处理实验		√	√		√						
机器人理论与技术实验				√	√			√			
程序设计训练		√	√		√						
数据结构实践		√	√		√						
软件工程课程设计			√							√	
软件开发技能实习			√							√	
认识实习		√				√					
生产实习				√		√		√		√	
毕业实习			√		√	√		√		√	
毕业设计(论文)			√	√	√	√			√		√
创新创业实践							√	√		√	
第二课堂								√	√		√
国家安全教育						√	√				
大学生劳动教育							√	√			

九、课程修读进程表

IX.Course Reading Progress Table



十三、教学环节设置及学分分布表

XI. Offered Course and Distribution of Academic Credits

课程类型			课程性质	课程编码	课程名称	学分	合计	课内学时			实践学时	学期	是否辅修/学位课程	先修课程/备注
								讲课	实验	上机				
平台	公共课	公共基础课程	必修	5103001	中国近现代史纲要 An Outline of Modern and Contemporary History of China	3	48	42	0	0	6	1		

程 平 台 课 程			5105017	思想道德与法治 Ideology and morality and rule of law	3	48	42	0	0	6	2		
			5101005	毛泽东思想和中国特色社会主义理论体系概论 Theoretical System of Socialism with Chinese Characteristics	3	48	32	0	0	16	3		
			5101006	习近平新时代中国特色社会主义思想概论 An Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	3	48	32	0	0	16	3		
			5102001	马克思主义基本原理 Fundamentals of Marxism	3	48	44	0	0	4	4		
			1401010	大学英语(一) College English (I)	2	32	32	0	0	0	1		
			1401062	大学英语(二) College English (II)	2	32	32	0	0	0	2		
			1401011	大学英语(三) College English (III)	2	32	32	0	0	0	3		
			1401064	大学英语(四) College English (IV)	1	16	16	0	0	0	4		
			1501882	体育(一) Physical Education(I)	1	26	26	0	0	0	1		
			1501883	体育(二) Physical Education(II)	1	34	34	0	0	0	2		
			1501884	体育(三) Physical Education(III)	1	34	34	0	0	0	3		
			1501885	体育(四) Physical Education(IV)	1	34	34	0	0	0	4		
			5106001	形势与政策 World Affairs and State Policy	2	64	64	0	0	0	1-8		分散进行
			1501024	体质健康测试 Physical Health Test	0	16	0	0	0	16	1-8		分散进行
	通 识 教 育 课 程	必修	2502006	大学生心理健康教育 Mental Health Education	2	32	24	0	0	8	1		
			2503001	职业生涯规划与就业指导 Career Planning and Employment Guidance	1	16	16	0	0	0	2		
			2504005	军事理论 Military Theory	2	36	36	0	0	0	2		

学科基础平台课程				8001003	大学生创新创业基础 Foundation of Innovation and Entrepreneurship for college students	1	16	16	0	0	0	2		
				5107001	国家安全教育 National Security Education	1	16	8	0	0	8	2		
				2501016	大学生劳动教育 College student labor education	1	32	2	0	0	30	1,2,3,4		
			选修	美育教育类 2 学分 Art Education2 Academic Credit										
				医学类 1 学分 Medicine1 Academic Credit										
				思想政治及新时代素质教育类 2 学分 Ideological Political Education2 Academic Credit										
				人文社科类 1 学分 Humanity Social Science1 Academic Credit										
				经济管理类 1 学分 Economic Management1 Academic Credit										
	大学 学科 基础 课程	必修	1303004	计算机科学导论 Introduction to Computer Science	1	16	16					1		
			1303068	C 语言程序设计 C Programming	3	48	48					1	是	
			0702601	高等数学 A(一) Advanced Mathematics A(I)	5	80	80	0	0	0		1		
		必修	0702602	高等数学 A(二) Advanced Mathematics A(II)	6.5	104	104	0	0	0		2		
			0703605	大学物理 B(一) College Physics B(I)	2.5	40	40	0	0	0		2		
			0703606	大学物理 B(二) College Physics B(II)	2	32	32	0	0	0		3		
			0703607	大学物理实验 B Experiments of College Physics B	1	24	0	24	0	0		3		
			0702103	线性代数 A Linear Algebra A	3	48	48	0	0	0		3		
			0702303	概率论与数理统计 A Probability and Mathematical Statistics(A)	3	48	48	0	0	0		4		
			1303096	离散数学 Discrete Mathematics	3	48	48	0	0	0		2		
			1303098	面向对象程序设计 Object-Oriented Programming	2.5	40	32	0	8	0		2		

		智能科学与技术专业必修课程	必修	1304065	Java 编程技术 Java Programming	2.5	40	30	0	1 0	0	3		
				1302133	数据结构 Data Structure	3.5	56	56	0	0	0	3		C 语 言 程 序 设 计
				1303017	数字逻辑与数字系统 Digital Logic and Digital System	3.5	56	46	10	0	0	3		
				1304095	人工智能 Artificial Intelligence	2	32	32	0	0	0	3		
				1302091	编译原理 Principles of Compiling	3	48	40	0	8	0	4		
				1302135	数据库系统原理 Principles of Database System	3	48	40	0	8	0	4		
				1303023	计算机组成原理 Principles of Computer Composition	3.5	56	56	0	0	0	4		数 字 逻 辑 与 数 字 系 统
				1307001	机器学习 Machine Learning	2	32	32	0	0	0	4		
				1302134	操作系统 Operating System	2.5	40	40	0	0	0	5		
				1304062	计算机网络 Computer Network	3.5	56	48	8	0	0	5		
				1350076	计算机视觉 Computer Vision	2.5	40	32	0	8	0	5		机 器 学 习
				1304094	软件工程 Software Engineering	2.5	40	32	0	8	0	5		
				1307002	自然语言处理 Nature Language Processing	2.5	40	40				6		机 器 学 习
				1307004	机器人理论与技术 Robotic Theory and Technology	2.5	40	40				6		
				1303105	算法设计与分析 Algorithm Design and Analysis	2.5	40	32	0	8	0	6		
		专 业 选 修 课程	选修	1304046	网络科学与工程导论 Introduction to Network Science and Engineering	2	32	32	0	0	0	2		
				1305030	网络空间安全导论 Introduction to Cybersecurity	2	32	32	0	0	0	2		
				1304092	Web 前端开发技术 Web Front End Development Technology	2	32	32	0	0	0	3		

				0702136	数学建模 Mathematical Modeling	2	32	24	0	8	0	5		
				1302144	数据库应用技术 Database Application Technology	2	32	28	0	4	0	5		
				1303106	移动应用开发 Development of Mobile Application	2	32	24	8	0	0	5		
				1303107	数值计算基础 Basics of Numerical Calculation	2	32	32	0	0	0	5		推荐 智能 科学与技 术专业
				1303108	图像处理 Image Processing	2	32	24	0	8	0	5		
				1304070	大数据技术原理与应用 Big Data Technology and Applications	2	32	26	0	6	0	5		
				1304090	知识图谱技术与应用 Knowledge graph technique and application	2	32	32	0	0	0	5		
				1305040	软件安全 Software Security	2	32	32	0	0	0	5		
				1302139	Java EE 架构 Java EE Architecture	2	32	32	0	0	0	6		
				1302140	数据挖掘 Data Mining	2	32	32	0	0	0	6		推荐 软件 工程专业
				1302145	Linux 内核与程序设计 Linux Kernel and Programming	2	32	32	0	0	0	6		推荐 软件 工程、 智能 科学与技 术专业
				1303095	虚拟现实与仿真 Virtual Reality and Simulation	2	32	24	8	0	0	6		

实践教学模块				1303102	接口与通讯技术 Interface and Communication Technology	2.5	40	34	6	0	0	6		
				1303109	专业英语 Subject-based English	2	32	32	0	0	0	6		
				1305017	信息内容安全 Information content Security	2	32	24	0	8	0	6		
				1304093	动手学深度学习 Hands-on Deep Learning	2	32	32	0	0	0	3		
	大 类 实 践 课 程	必修		1303008	C 语言程序设计实验 Experiments in C Programming	1	24	0	0	2 4	0	1		
				2504006	军事训练 Military Training	2	112	0	0	0	112	1		
	智能 科 学 与 技 术 专 业	必修		1307005	自然语言处理实验 Experiments in Nature Language Processing	1	24		24		24	6		
				1307006	机器人理论与技术实验 Experiments in robotic theory and technology	1	24		24		24	6		
				1305025	认识实习 Cognition Practice	1	1(周)	0	0	0	1(周)	4		
				1303015	程序设计训练 Advanced Training for Programming Design	1	1(周)	0	1(周)	0	0	3		
				1302006	数据结构实践 Data Structure Practice	1	1(周)	0	1(周)	0	0	4		
				1302034	数据结构课程设计 Course Project in Data Structure	1	1(周)	0	0	0	1(周)	4		
				1303009	计算机组成原理实验 Experiments in Principles of Computer Organization	1	24	0	24	0	0	4		
				1302132	数据库系统课程设计 Course Project in Database System	1	1(周)	0	0	0	1(周)	5		
				1303036	软件开发技能实习 Practice Experience in Software Development Skills	2	2(周)	0	0	0	2(周)	5		
				1302122	操作系统课程设计 Course Project in Operating System	1	1(周)	0	0	0	1(周)	6		

				1302131	软件工程课程设计 Course Project in Software Engineering	1	1(周)	0	0	0	1(周)	6		
				1304022	计算机网络课程设计 Course Project in Computer Network	1	1(周)	0	0	0	1(周)	6		
				1303097	生产实习 Production Practice	2	2(周)	0	0	0	2(周)	6		
				1303055	毕业实习 Pre-graduation Internship	2	2(周)	0	0	0	2(周)	7		
				1303057	毕业设计(论文) Undergraduate Project (Thesis)	14	14(周)	0	0	0	14(周)	8		
	素质拓展模块	创新创业教育	必修		创新创业实践 3 学分 Innovation Practices 3 Academic Credits									
		第二课堂			第二课堂 3 学分 Second Classroom 3 Academic Credits									